CHRISTIAN OMAR PAYÁN TORRÓNTEGUI

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SUMMARY

Robotics & Digital Systems Engineering student (8th Semester) specializing in 3D printing, additive manufacturing, and rapid physical prototyping. Proven experience operating FDM/SLA 3D printers, designing CAD assemblies, and optimizing prototyping costs (80–95% cost reductions). Passionate about supporting innovation labs, ideation workshops, and hardware-software validation for next-generation mobility technology.

EDUCATION

B.S. Robotics and Digital Systems Engineering (8th Semester) Graduating Expected Dec 2026
Tecnológico de Monterrey
Zapopan, Jalisco
Relevant coursework: Product Design & 3D Modeling (Rhino 3D, SOLIDWORKS), Additive Manufacturing, Rapid Prototyping, Embedded Systems, Agile/Scrum

TECHNICAL SKILLS

Additive Manufacturing & 3D Printing: FDM, SLA, SLS, Slicing Optimization (Cura, PrusaSlicer), Polymer Selection (PETG, TPU, PLA), Rapid Prototyping
Product & CAD Design: SOLIDWORKS, Fusion 360, Rhino 3D, CAD/CAM, Tolerancing, Assembly Design, BOM Management
Systems & Electronics: Custom PCB Design, STM32, ESP32, Embedded Hardware Integration, Sensors, Microcontrollers
Languages & Methods: English (Advanced / B2 Professional), Spanish (Native), Git, Linux (Ubuntu), Agile/Scrum, Ideation Workshops

PROFESSIONAL EXPERIENCE

Industrial Mounts & Hardware Interfaces, Zapopan, Jalisco: Mechanical & Prototyping Designer Feb - Jun 2026

- Engineered ergonomic hardware interfaces and 3-DOF mechanical mounts, reducing prototyping costs by 95% vs commercial alternatives
- Operates 3D printers (FDM/PETG) for rapid enclosure iteration, verifying thermal, mechanical, and assembly tolerances for robust physical deployment
- Documented BOM structures and continuous improvement iterations during rapid product prototyping phases

ACADEMIC & TECHNICAL PROJECTS

Open-Source Medical Phantom — Additive Manufacturing Aug 2025 - Jun 2026
FDM 3D Printing, TPU/PETG, Rapid Prototyping

- Engineered anatomical testing phantom using advanced FDM 3D printing, achieving an 80% cost reduction vs commercial trainers
- Executed rapid physical design iterations under strict material constraints, quality requirements, and dimensional safety limits

Embedded Systems Hardware Integration & Custom Enclosures Aug - Dec 2024
Custom PCB, 3D Printed Enclosures, Microcontrollers

- Modeled and 3D-printed custom multi-component enclosures for embedded system hardware with integrated sensor cutouts
- Coordinated end-to-end prototyping workflow under Scrum methodology with zero mechanical fit errors

Autonomous Mobile Robot Sensor Mounts & Hardware Design Aug - Dec 2025
3D Printing, ROS 2, URDF, Jetson, Optical Sensors

- Designed and 3D-printed custom mechanical mounts and protective housings for LiDAR, cameras, and onboard compute units
- Validated physical structural integrity and sensor alignment under real-world vibration and mechanical strain